

Project Report: Enhancing Real Estate Valuation via Multimodal Learning and Feature Extraction

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1. Executive Summary

This report details the iterative development of a property valuation system that integrates tabular housing data with satellite imagery. Moving beyond standard Automated Valuation Models (AVMs), we explored a progression of architectures, from a tabular baseline to complex hybrid neural networks. A key breakthrough involved using Grad-CAM to interpret model focus, which motivated a pivot toward explicit feature extraction using OpenCV. The final deployed solution utilizes an ensemble approach, combining the stability of the baseline model with the contextual nuances of the visual model.

2. Exploratory Data Analysis (EDA) & Preprocessing

While the primary focus of this study was architectural, two critical data observations dictated our preprocessing strategy:

- **Target Variable Skewness:** The raw property price distribution exhibited significant right-skewness. To address this and stabilize the variance of the residuals, we applied a **Log Transformation** ($\log(1+x)$) to the target variable. This allowed the models to minimize Mean Squared Error (MSE) in a normalized space rather than over-penalizing high-value outliers.
- **The Geospatial Non-Linearity:** The correlation matrix initially suggested a weak linear relationship between coordinates (**Latitude**, **Longitude**) and **Price**. However, this was a misleading metric. Spatial plotting revealed distinct, non-linear high-value clusters. This confirmed that while location is not linearly correlated (i.e., moving "North" doesn't always increase price), it is spatially deterministic, requiring non-linear models (like Tree-based methods or Neural Networks) to capture these boundaries.

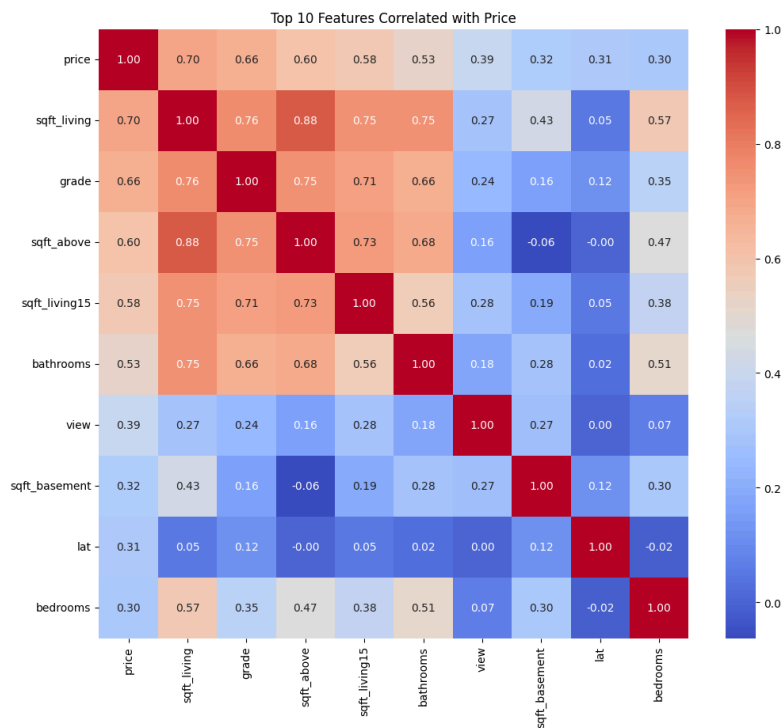


Fig: Correlation Matrix

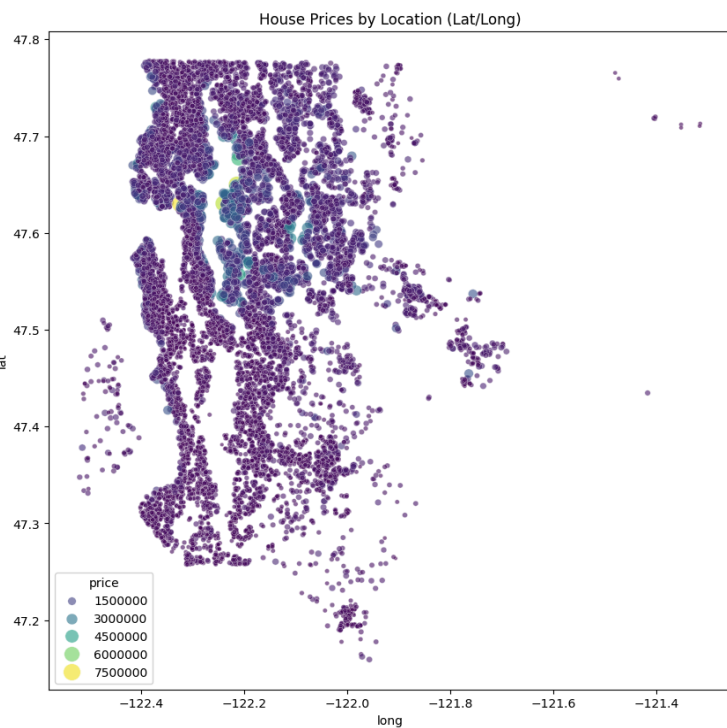


Fig: locations (lat vs long) marked

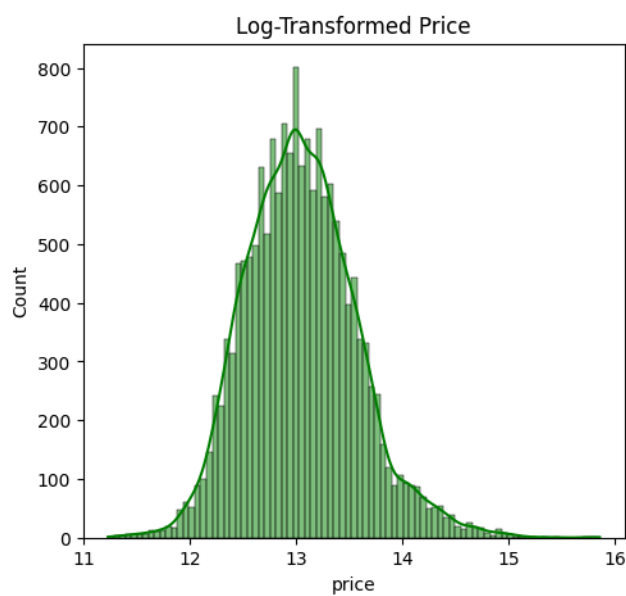
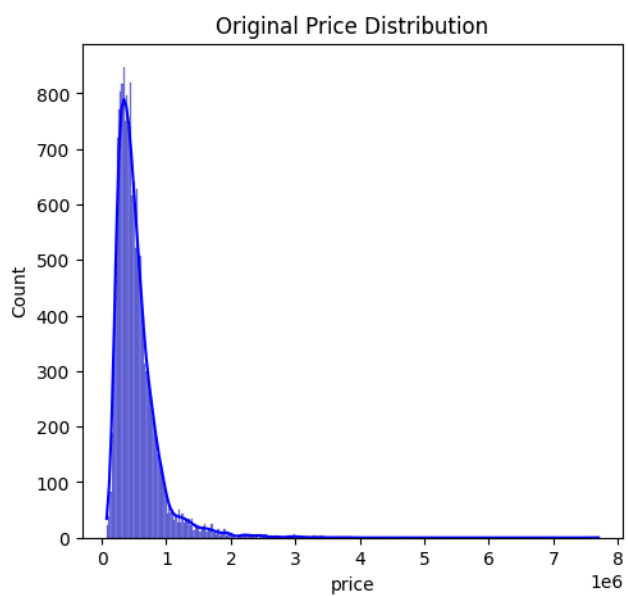


Fig: Log-Transformation done for highly 'right' skewed price distribution

3. Modeling Evolution

Our modeling phase followed a strict iterative path, moving from simple interpretability to complex multimodal integration.

3.1 The Baseline Model (Tabular Only)

We established a benchmark using a Gradient Boosting Regressor (XGBoost) trained exclusively on structural features (Bedrooms, Bathrooms, SqFt) and location. We got a R^2 of **0.8862**.

- **Pros:** Highly interpretable, fast training, and robust against noise.
- **Cons:** It operated in a "blind" environment, treating a property next to a landfill the same as one next to a park if their structural stats were identical. This resulted in an irreducible error floor.

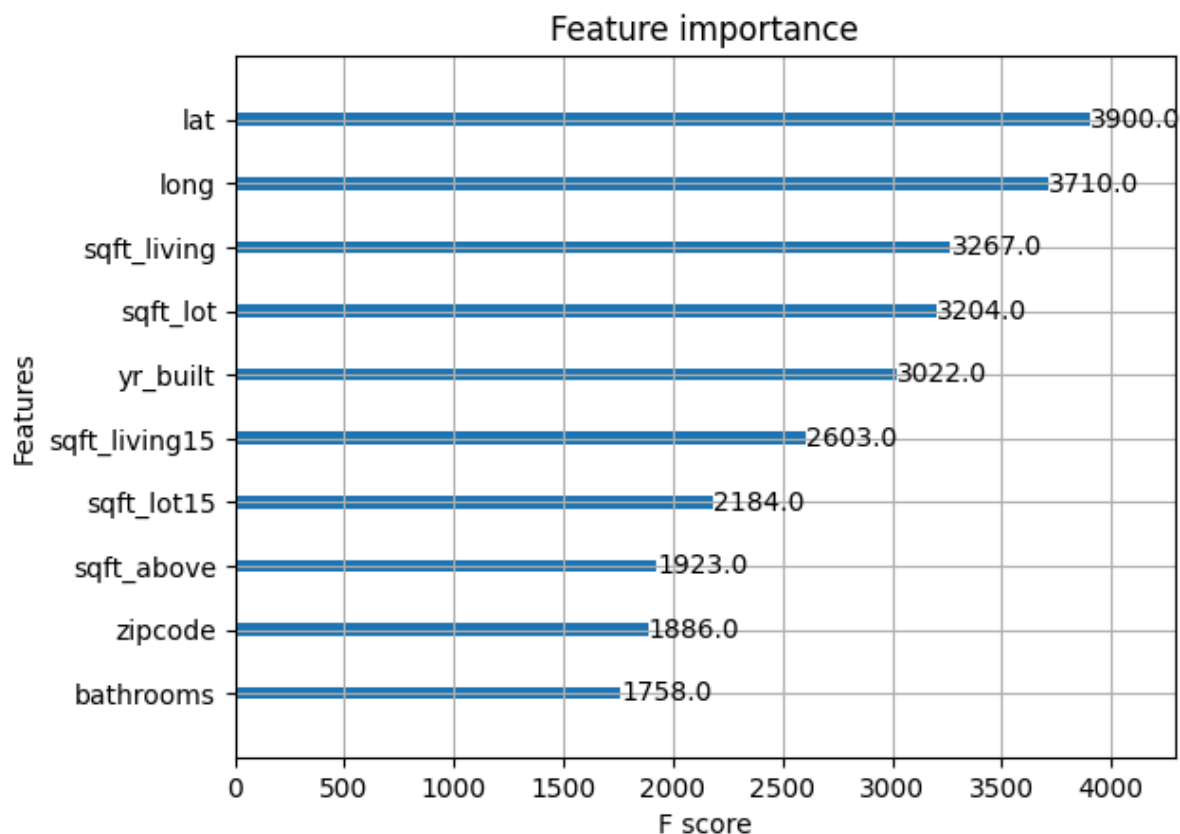


Fig: "Location! Location! And Location!", clearly our model believes that location of a property is most influential factor on its price

3.2 The Hybrid Model (Early Fusion)

We introduced satellite imagery by passing images through a Convolutional Neural Network (CNN) backbone. The flattened image embeddings were concatenated directly with the tabular feature vector before the final regression layers.

- **Performance:** Slightly improved over baseline but suffered from high variance.
- **Failure Point:** The "curse of dimensionality." The high-dimensional image vectors often overwhelm the sparse tabular signal, causing the model to overfit to visual noise (e.g., shadows or cars in driveways) rather than property value.

3.3 The Gated Fusion Model

To counter the noise issue in the Hybrid model, we implemented a **Gated Multimodal Unit**. This architecture utilized a learnable "gate" (sigmoid activation) to dynamically weight the importance of the visual features.

- **Hypothesis:** The model would learn to ignore the image if the satellite view was occluded or irrelevant.
- **Outcome:** While theoretically sound, the model struggled to converge. The gating mechanism added optimization complexity without yielding a statistically significant improvement in RMSE compared to the simpler Hybrid approach.

4. Visual Interpretability & The Feature Engineering Pivot

4.1 Grad-CAM Analysis

To understand *why* our Deep Learning models were underperforming or behaving erratically, we employed **Grad-CAM (Gradient-weighted Class Activation Mapping)**. This technique visualized the specific regions of the satellite images that triggered the model's activation.

- **The Insight:** The heatmaps revealed a critical flaw: the Deep Learning models were often focusing on irrelevant artifacts, such as clouds, or the brightness of the street, rather than the house itself.
- **The Pivot:** However, in successful predictions, Grad-CAM showed the model focusing on **vegetation (greenery)** and **amenities (pools, paved driveways)**.

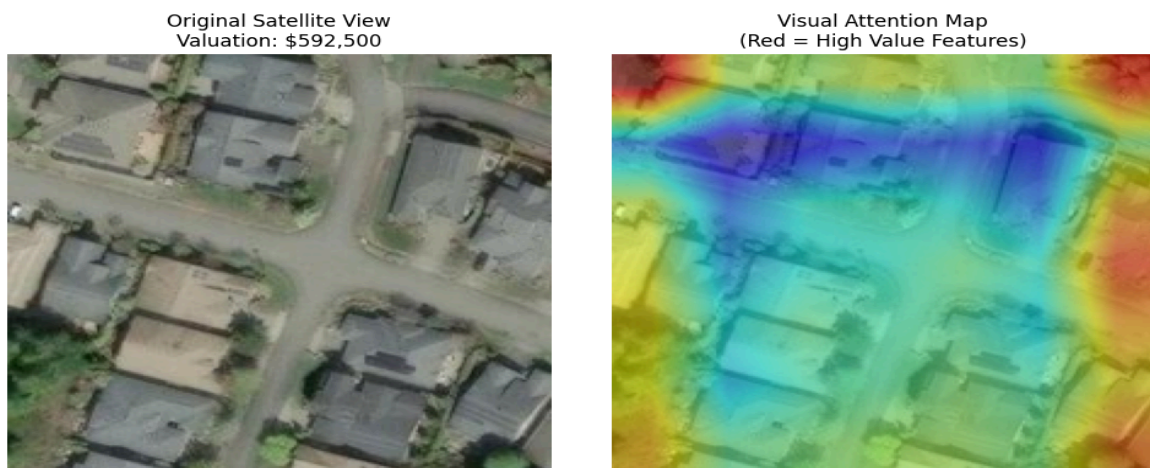


Fig: Our model gives some weight to factors such as “roof” and “greenery”.

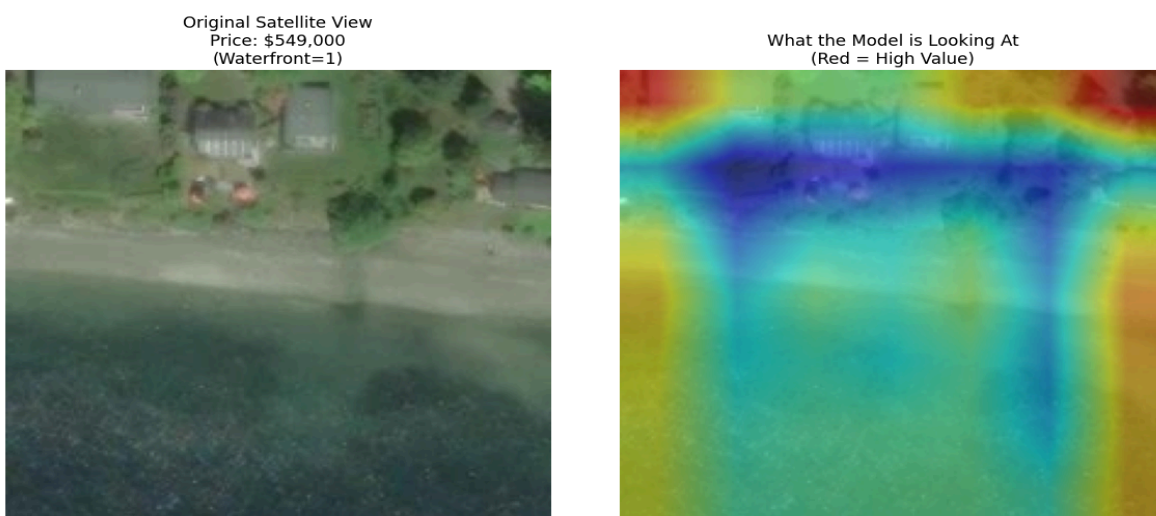


Fig: Model isn't able to give importance to “water”. This shows that our model is still quite weak in identifying important areas.

4.2 Explicit Feature Extraction (OpenCV)

Instead of hoping the "Black Box" CNN would learn these features correctly, we decided to **manually extract** them using Computer Vision techniques (OpenCV):

- **Greenery Index:** We calculated the percentage of green pixels (HSV masking) to quantify "curb appeal" and neighborhood health.
- **Structure Detection:** We applied edge detection and contouring to estimate the building-to-land ratio.
- **Why this worked:** This replaced thousands of noisy raw pixel values with a few high-signal, engineered features. It eliminated the noise that was confusing the Neural Networks.

Using this architecture, we got an R^2 of **0.8899!**

5. Final Architecture: The Ensemble Approach

The final deployed solution was neither the Baseline nor the Deep Learning model alone, but a **Weighted Ensemble**.

The Logic

We discovered that the **Baseline Model** was superior for "standard" houses where structural data is the primary driver of price. The **Image/OpenCV Enhanced Model** excelled in "outlier" cases where extrinsic factors (e.g., a massive garden, a corner plot, or proximity to water) justified a premium that tabular data couldn't explain.

Conclusion

By averaging the predictions of the Baseline and the Feature-Enhanced model, we achieved a lower error rate than any single model could achieve in isolation. This project demonstrated that while Deep Learning provides powerful capabilities, **domain-specific feature engineering** guided by interpretability tools (Grad-CAM) remains the gold standard for high-stakes regression tasks.